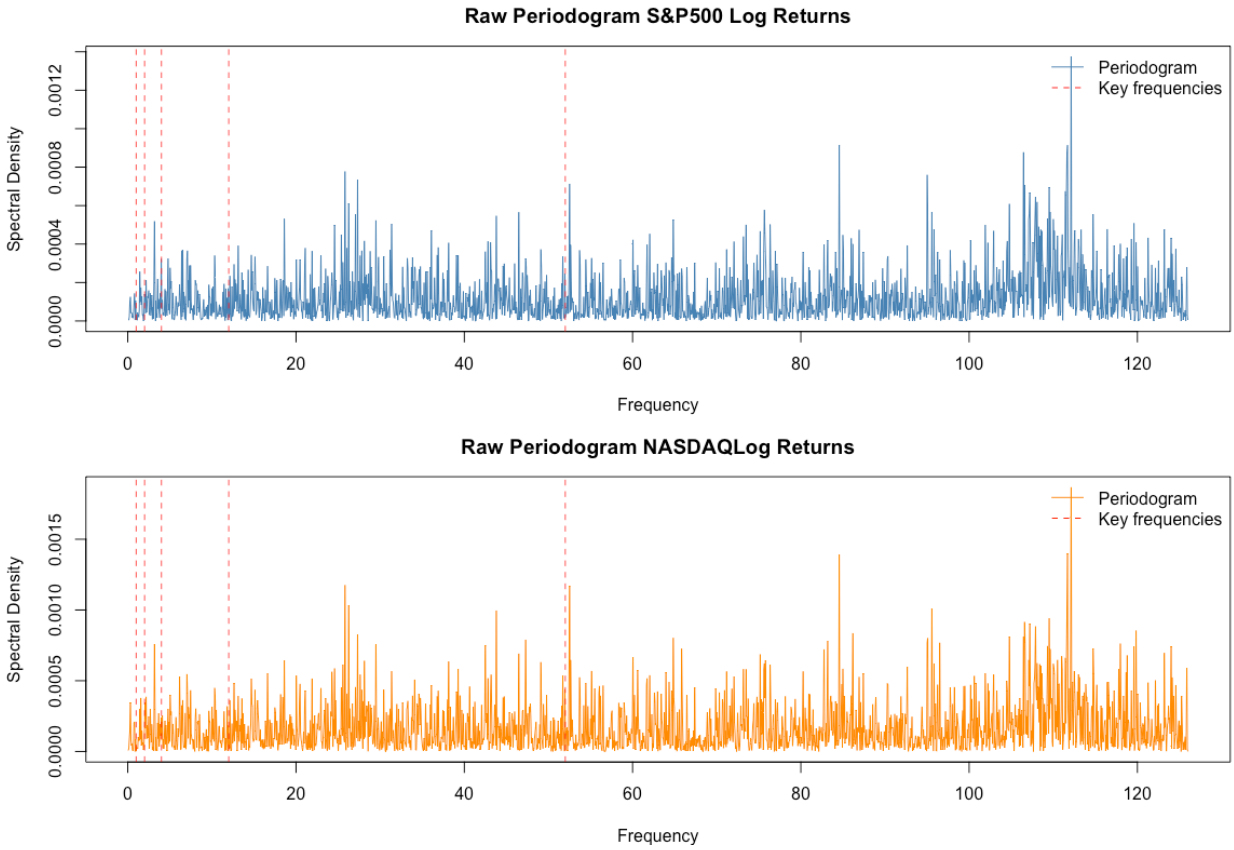
STAT-S 650 Time Series Analysis

Assignment 4

Manish Suresh Maudgalya

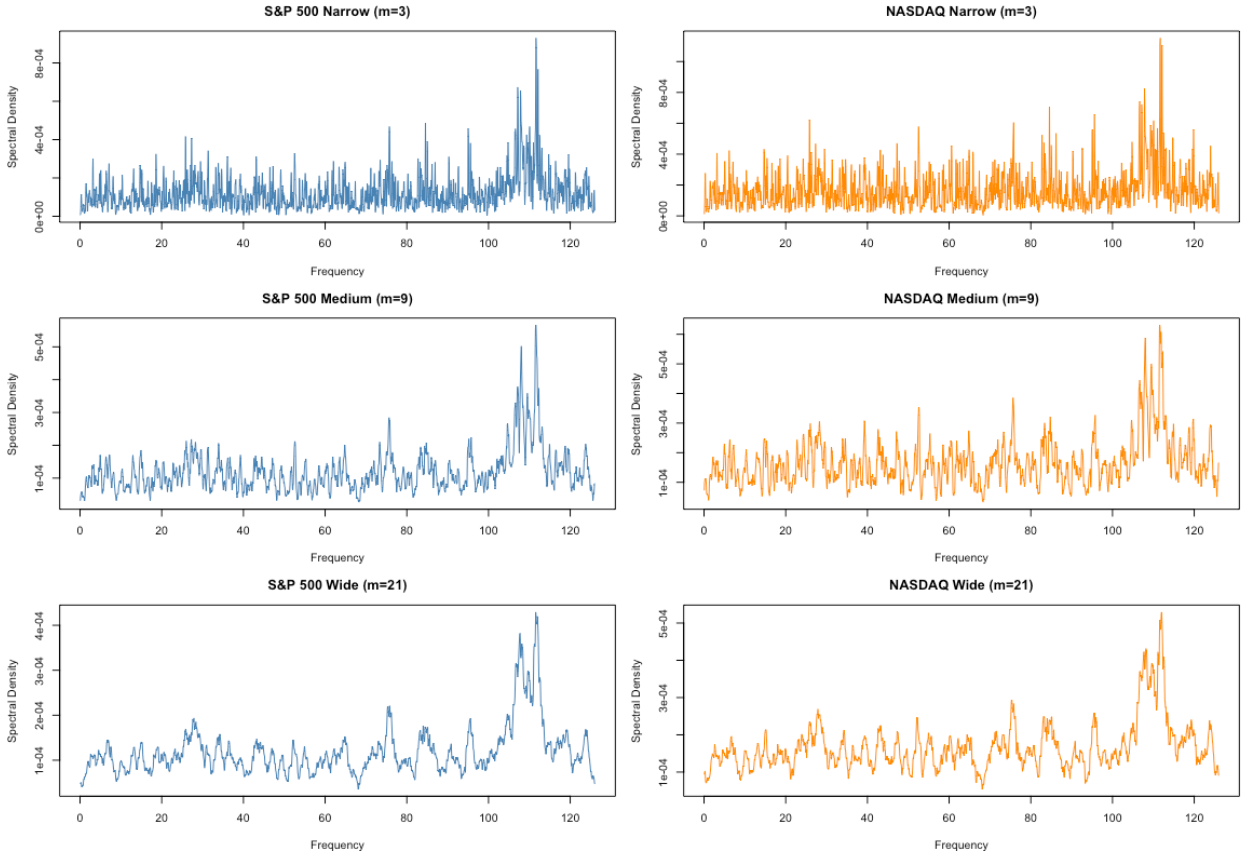
Q. Compute and plot the periodogram separately for each measure. Do you have any systematic cycles that stick out extremely large. If so, use regression to remove them.

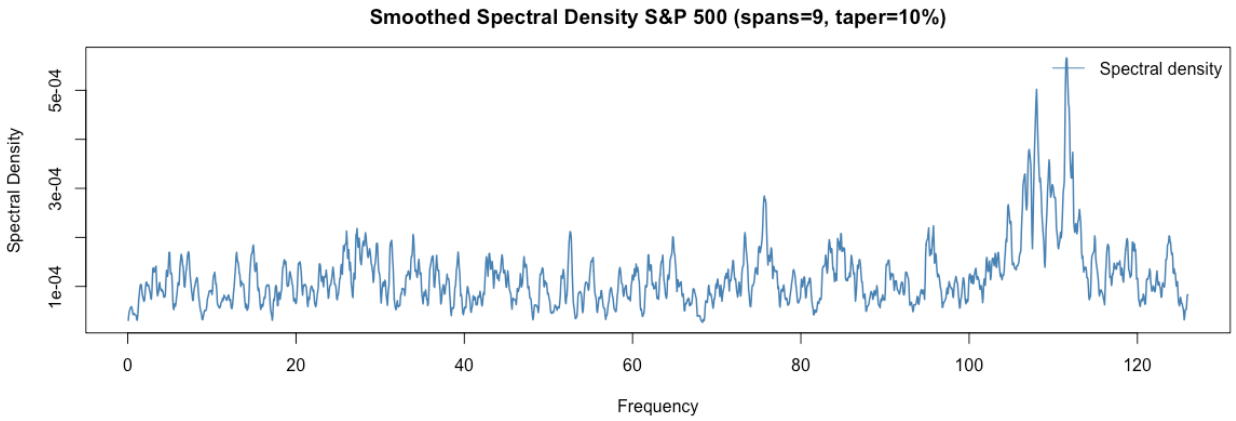


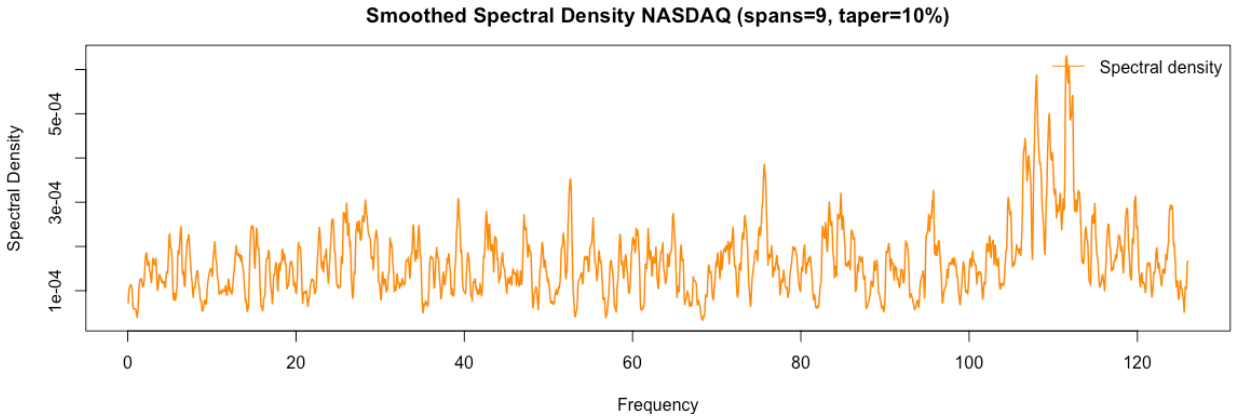
Periodogram Result

- Both the S&P 500 and NASDAQ raw periodograms display a broadly flat spectrum with power distributed relatively evenly across all frequencies, which is characteristic of a possible white noise process.
- No single frequency exhibits a dominant spike that would indicate a systematic cycle. The isolated tall spikes visible (particularly in the NASDAQ around frequencies 85–115) are possibly due to the sampling in the periodogram.
- There is no real seasonal cycle in either the SP or Nasdaq log returns.

Q. Compute and plot the estimated spectral distribution separately for each measure. Try out different window sizes. State which size you used and why you chose this method? Interpret the spectral results.

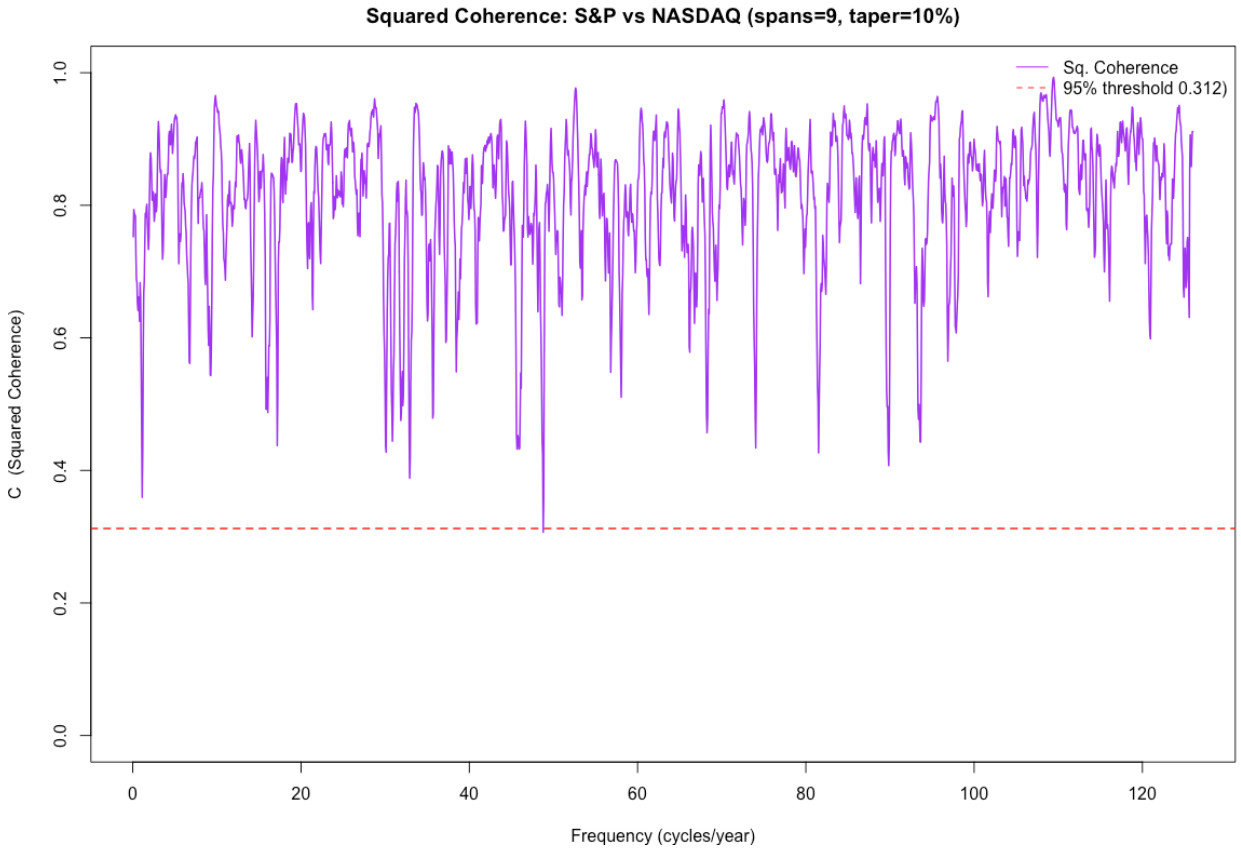






- The narrow window (m=3) was too noisy, the wide window (m=21) over smoothed and merged features. So, I chose window size of 9.
- Both spectra are broadly flat across low and mid frequencies, meaning there are no dominant seasonal or long run cyclical components in the log returns.
- A peak at high frequencies around 105–115 is seen in both indices across all window sizes, possibly a genuine short term cyclical pattern due to weekly some trading spike.
- The almost identical spectral shapes of SP 500 and NASDAQ across all window sizes show that the two series share the same underlying frequency structure

Q. Compute the coherence between the two measures. Interpret these results.



- Squared coherence stays well above the 95% threshold (0.312) across almost the entire frequency range could mean that the two indices move together significantly almost always.
- Values consistently sit between 0.6 and 0.95, which is high meaning that any cyclical pattern that exists in one index the other follows too.
- The bottom line is that S&P 500 and NASDAQ are tightly coupled in the frequency domain.

R Code

```r
# Spectral Analysis. S&P 500 vs NASDAQ

library(ggplot2)
library(dplyr)
library(tidyr)
library(astsa)
df <- read.csv("/Users/manishsm/Desktop/Spring 2026/Time Series Analysis/Assignments/quantmod_dataset.csv")
df$Date <- as.Date(df$Date)
df <- df[order(df$Date), ]

# log returns
gspc <- diff(log(df$GSPC.Adjusted))
ixic <- diff(log(df$IXIC.Adjusted))

n <- length(gspc)
cat("Number of observations of log returns:", n, "\n")

# Periodograms
pg_gspc <- spec.pgram(gspc, taper = 0, log = "no", plot = FALSE)
pg_ixic <- spec.pgram(ixic, taper = 0, log = "no", plot = FALSE)

freq_year_gspc <- pg_gspc$freq * 252
freq_year_ixic <- pg_ixic$freq * 252


par(mfrow = c(2, 1), mar = c(4, 4, 3, 1))

plot(freq_year_gspc, pg_gspc$spec,
     type = "l", col = "steelblue",
     xlab = "Frequency",
     ylab = "Spectral Density",
     main = "Raw Periodogram S&P500 Log Returns")
abline(v = c(1, 2, 4, 12, 52, 252), col = "red", lty = 2, lwd = 0.8)
legend("topright",
       legend = c("Periodogram", "Key frequencies"),
       col    = c("steelblue", "red"),
       lty    = c(1, 2), bty = "n")

plot(freq_year_ixic, pg_ixic$spec,
     type = "l", col = "darkorange",
     xlab = "Frequency",
     ylab = "Spectral Density",
     main = "Raw Periodogram NASDAQLog Returns")
abline(v = c(1, 2, 4, 12, 52, 252), col = "red", lty = 2, lwd = 0.8)
legend("topright",
       legend = c("Periodogram", "Key frequencies"),
       col    = c("darkorange", "red"),
       lty    = c(1, 2), bty = "n")

```

```r
# Checking Cycles
top_gspc <- order(pg_gspc$spec, decreasing = TRUE)[1:5]
top_ixic <- order(pg_ixic$spec, decreasing = TRUE)[1:5]

cat("\nTop 5 dominant frequencies S&P 500in cycles/year:\n")
print(data.frame(
  freq_per_year = round(pg_gspc$freq[top_gspc] * 252, 3),
  period_days   = round(1 / pg_gspc$freq[top_gspc], 1),
  power         = round(pg_gspc$spec[top_gspc], 6)
))

cat("\nTop 5 dominant frequencies NASDAQ in cycles/year:\n")
print(data.frame(
  freq_per_year = round(pg_ixic$freq[top_ixic] * 252, 3),
  period_days   = round(1 / pg_ixic$freq[top_ixic], 1),
  power         = round(pg_ixic$spec[top_ixic], 6)
))


remove_cycle <- function(x, freq_dominant) {
  t <- seq_along(x)
  cos_term <- cos(2 * pi * freq_dominant * t)
  sin_term <- sin(2 * pi * freq_dominant * t)
  fit <- lm(x ~ cos_term + sin_term)
  cat("Regression summary for cycle removal:\n")
  print(summary(fit))
  residuals(fit)
}

# gspc_detrended <- remove_cycle(gspc, freq_dominant =
pg_gspc$freq[top_gspc[1]])
# ixic_detrended <- remove_cycle(ixic, freq_dominant =
pg_ixic$freq[top_ixic[1]])

# 4 Smoothed Spectral Density trying with different windw sizes
par(mfrow = c(3, 2), mar = c(4, 4, 3, 1))

spans_list <- list(
  "Narrow (m=3)"  = c(3),
  "Medium (m=9)"  = c(9),
  "Wide (m=21)"   = c(21)
)

for (nm in names(spans_list)) {
  sp <- spans_list[[nm]]

  sm_gspc <- spec.pgram(gspc, spans = sp, taper = 0.1, log = "no", plot =
FALSE)
  sm_ixic <- spec.pgram(ixic, spans = sp, taper = 0.1, log = "no", plot =
FALSE)

  plot(sm_gspc$freq * 252, sm_gspc$spec,
       type = "l", col = "steelblue",
       xlab = "Frequency (cycles/year)", ylab = "Spectral Density",
```

```r
       main = paste("GSPC -", nm))
  abline(v = c(1, 5, 52), col = "red", lty = 2, lwd = 0.8)

  plot(sm_ixic$freq * 252, sm_ixic$spec,
       type = "l", col = "darkorange",
       xlab = "Frequency (cycles/year)", ylab = "Spectral Density",
       main = paste("IXIC -", nm))
  abline(v = c(1, 5, 52), col = "red", lty = 2, lwd = 0.8)
}

#CHOSEN window 9
cat("\nFinal chosen spectral estimates (spans = 9, taper = 10%)\n")
final_gspc <- spec.pgram(gspc, spans = 9, taper = 0.1, log = "no", plot = FALSE)
final_ixic <- spec.pgram(ixic, spans = 9, taper = 0.1, log = "no", plot = FALSE)

par(mfrow = c(2, 1), mar = c(4, 4, 3, 1))

plot(final_gspc$freq * 252, final_gspc$spec,
     type = "l", col = "steelblue", lwd = 1.5,
     xlab = "Frequency (cycles/year)", ylab = "Spectral Density",
     main = "Smoothed Spectral Density - GSPC (spans=9, taper=10%)")
abline(v = c(1, 5, 12, 52), col = "red", lty = 2)
legend("topright",
       legend = c("Spectral density", "Key annual freqs"),
       col = c("steelblue", "red"), lty = c(1, 2), bty = "n")

plot(final_ixic$freq * 252, final_ixic$spec,
     type = "l", col = "darkorange", lwd = 1.5,
     xlab = "Frequency (cycles/year)", ylab = "Spectral Density",
     main = "Smoothed Spectral Density - IXIC (spans=9, taper=10%)")
abline(v = c(1, 5, 12, 52), col = "red", lty = 2)
legend("topright",
       legend = c("Spectral density", "Key annual freqs"),
       col = c("darkorange", "red"), lty = c(1, 2), bty = "n")

# 5 Coherence Between GSPC and IXIC
par(mfrow = c(1, 1), mar = c(4, 4, 3, 1))

coh <- spec.pgram(cbind(gspc, ixic),
                  spans   = 9,
                  taper   = 0.1,
                  log     = "no",
                  plot    = FALSE)

# Squared coherence
sq_coh <- Mod(coh$coh)^2

plot(coh$freq * 252, sq_coh,
     type = "l", col = "purple", lwd = 1.5,
     ylim = c(0, 1),
     xlab = "Frequency (cycles/year)",
     ylab = expression(hat(C)^2 ~ "(Squared Coherence)"),
```

```r
     main = "Squared Coherence: GSPC vs IXIC (spans=9, taper=10%)")


m_df <- length(spans_list[["Medium (m=9)"]])
L <- 9
threshold_95 <- 1 - (0.05)^(1 / (L - 1))
abline(h = threshold_95, col = "red", lty = 2, lwd = 1.5)
legend("topright",
       legend = c("Sq. Coherence", paste0("95% threshold ",
round(threshold_95, 3), ")")),
       col = c("purple", "red"), lty = c(1, 2), bty = "n")

cat("\n95% coherence significance threshold:", round(threshold_95, 4), "\n")
cat("Mean squared coherence across frequencies:", round(mean(sq_coh), 4),
"\n")
cat("Frequencies (cycles/year) where coherence > threshold:\n")
sig_freqs <- coh$freq[sq_coh > threshold_95] * 252
cat(round(range(sig_freqs), 2), "(range)\n")
cat("Proportion of frequencies with significant coherence:",
    round(mean(sq_coh > threshold_95), 3), "\n")
```

STAT-S 650 HOMEWORK 4

MANISH
SURESH MANDLALYA

Theory

1). Given $Y(t) = \alpha \cdot Y(t-1) + W(t) + \beta \cdot W(t-1)$

Given spectral density of ARMA from textbook.

$$f_x(w) = \frac{\sigma_w^2 \left| \theta(e^{-2\pi i w}) \right|_2}{\left| \phi(e^{-2\pi i w}) \right|^2}$$

Where $\phi(z) = 1 - \sum_{k=1}^{p} \phi_k z^k$

$$\theta(z) = 1 + \sum_{k=1}^{q} \theta_k z^k$$

$W(t)$ is white noise.

-> lets assume $BY(t) = Y(t-1)$

$B W(t) = W(t-1)$.

$$Y(t) - \alpha B Y(t) = (1 + \beta B) W(t)$$

$$\Rightarrow (1 - \alpha B) Y(t) = (1 + \beta B) W(t).$$

$\Rightarrow$ Transfer function is $H(B) = \frac{1 + \beta B}{1 - \alpha B}$

Using the property

$$f_Y(\omega) = \sigma_w^2 \frac{\theta |e^{-2\pi i \omega}|^2}{\phi |e^{-2\pi i \omega}|^2}$$

$$\Rightarrow \theta(e^{-2\pi i \omega}) = 1 + \beta e^{-2\pi i \omega}$$
$$\phi(e^{-2\pi i \omega}) = 1 - \alpha e^{-2\pi i \omega}.$$

$$f_Y(\omega) = \sigma_w^2 \frac{|1+\beta e^{-2\pi i \omega}|^2}{|1-\alpha e^{-2\pi i \omega}|^2}$$

$$\Rightarrow |1+\beta e^{-2\pi i \omega}|^2 = (1+\beta e^{-2\pi i \omega})(1+\beta e^{2\pi i \omega})$$
$$= 1 + \beta^2 + 2\beta \cos(2\pi\omega)$$

$$|1-\alpha e^{-2\pi i \omega}|^2 = (1-\alpha e^{-2\pi i \omega})(1-\alpha e^{2\pi i \omega}) = 1 + \alpha^2 - 2\alpha\cos(2\pi\omega)$$

∴ Final spectral density

$$f_Y(\omega) = \sigma_w^2 \frac{1 + \beta^2 + 2\beta\cos(2\pi\omega)}{1 + \alpha^2 - 2\alpha\cos(2\pi\omega)}$$

Hence proved.

2) AR (2) process $w(t)$ is noise

given

$$y(t) = \alpha_1 \cdot y(t-1) + \alpha_2 \cdot y(t-2) + w(t)$$

→ lets assume L is the lag operator

$$Ly(t) = y(t-1)$$

AR(2) model becomes.

$$(1 - \alpha_1 L - \alpha_2 L^2)\, y(t) = w(t)$$

→ $\phi(L) = 1 - \alpha_1 L - \alpha_2 L^2$.

$$y(t) = \frac{1}{\phi(L)}\, w(t).$$

Replacing L with $e^{-i\omega}$

$$\phi(e^{-i\omega}) = 1 - \alpha_1 e^{-i\omega} - \alpha_2 e^{-i2\omega}.$$

→ Power Spectrum

$$S_y(\omega) = \frac{\sigma_w^2}{|\phi(e^{-i\omega})|^2} = \frac{\sigma_w^2}{|1 - \alpha_1 e^{-i\omega} - \alpha_2 e^{-i2\omega}|^2}$$

$$|\phi(e^{-i\omega})|^2 = \phi(e^{i\omega}) \cdot \phi(e^{i\omega}).$$

$$= 1 + \alpha_1^2 + \alpha_2^2 - 2\alpha_1(1+\alpha_2)\cos\omega - 2\alpha_2 \cos(2\omega)$$

∴ Final Spectrum of AR(2) → $\dfrac{\sigma_w^2}{1 + \alpha_1^2 + \alpha_2^2 - 2\alpha_1(1+\alpha_2)\cos\omega - 2\alpha_2 \cos(2\omega)}$